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Appraisal Considered as a Process of Multilevel Sequential Checking

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This chapter describes the *sequential check theory of emotion differentiation* as part of a dynamic model of emotion (the *component process model of emotion*). The theory attempts to explain the differentiation of emotional states as the results of a sequence of specified stimulus evaluation (appraisal) checks and makes predictions concerning the ensuing response patterning in several organismic subsystems. Given that this book reviews the appraisal theory approach in general, together with chapters by major contributors to this tradition, this chapter will focus exclusively on sequential check theory. Comparative reviews have been published elsewhere (Scherer, 1988b; 1999a; see also Roseman & Smith; Schorr [a]; this volume). Preliminary versions of parts of this model have appeared in conference proceedings, book chapters, and empirical papers (Scherer, 1981b, 1982a, 1984a, 1984c, 1986b, 1988a, 1992c, 1993b, 1997a, 1999a, 1999b, 2000b). The most complete description of the model (Scherer, 1987a), while widely distributed, has never been formally published.[1] In the course of the development of the theory, details of the predictions as well as some aspects of the terminology have evolved. In this chapter, a systematic description of the most recent version of the theory, including detailed predictions and a review of the available evidence, is presented.[2]

The Component Process Model

Following similar suggestions in the literature, I have described emotion as an evolved, phylogenetically continuous mechanism that allows increasingly flexible adaptation to environmental contingencies by decoupling stimulus and response and thus creating a latency time for response optimization (Scherer, 1979, 1981b, 1984c, 1987a). As in other areas of organismic functioning, emotion interacts with phylogenetically older response mechanisms such as reflexes and fixed action patterns. Emotion is considered to be a theoretical construct that consists of five components corresponding to five distinctive functions (see table 5.1 for a list of the functions, the systems that subserve them, and the respective emotion components).[3] The theoretical analysis presented here assumes a continuously operating evaluation (appraisal) process and suggests that the functionally defined organismic subsystems (and thus the components of emotion) are multiply and recursively interrelated (that is, changes in one component can lead directly to corresponding changes in others).

Table 5.1. Relationships between the functions and components of emotion and the organismic subsystems that subserve them

Emotion function	Emotion component	Organismic subsystem (and major substrata)
Evaluation of objects and events	Cognitive component	Information processing (CNS)
System regulation	Peripheral efference component	Support (CNS, NES, ANS)
Preparation and direction of action	Motivational component	Executive (CNS)
Communication of reaction and behavioral intention	Motor expression component	Action (SNS)
Monitoring of internal state and organism-environment interaction	Subjective feeling component	Monitor (CNS)

CNS: central nervous system; NES: neuro-endocrine system; ANS: autonomic nervous system; SNS: somatic nervous system. The organismic subsystems are theoretically postulated functional units or networks.

In the framework of the component process model, emotion is defined as *an episode of interrelated, synchronized changes in the states of all or most of the five organismic subsystems in response to the evaluation of an external or internal stimulus event as relevant to major concerns of the organism*. In other words, it is suggested to use the term “emotion” only for those periods of time during which many organismic subsystems are coupled or synchronized to produce an adaptive reaction to an event that is considered as central to the individual’s well-being. The major features of this definition are discussed in greater detail in Scherer (1987a, 1993b) as well as in a recent attempt to use the concepts of nonlinear dynamics to define emotion (Scherer, 2000b).[4]

The Sequential Check Theory of Emotion Differentiation

Almost all theories of emotion assume, at least implicitly, that the specific kind of emotion experienced depends on the result of an evaluation or appraisal of an event in terms of its significance for the survival and well-being of the organism. The nature of this evaluation process has been rarely specified, even though many philosophers had shown the way by identifying some of the major dimensions inherent in the evaluation of the significance of events (Gardiner, Clark-Metcalf, & Beebe-Center, 1937). Arnold (1960a) and Lazarus (1966) were the first emotion theorists to attempt a more explicit description of the appraisal process (see chapters on the history and general approach of appraisal theories by Schorr [a] and Roseman & Smith, this volume). Building on these earlier approaches and on the observation that the valence, activation, and power dimensions of emotional meaning seemed to be linked to criteria of stimulus evaluation (valence = goal/need conduciveness, activation = urgency, and power = coping potential), I originally suggested a set of criteria (which I called *stimulus evaluation checks*, SECs), that are predicted to underlie the assessment of the significance of a stimulus event for an organism (Scherer, 1981b, 1984a, 1984c).[5]

While the number and definition of these SECs has evolved with the development of the theory, the underlying principle for theory building has remained constant. The SECs are chosen, in a principled fashion, to represent the mininal set of dimensions or criteria that are considered necessary to account for the differentiation of the major families of emotional states (see Scherer, 1997a, for a more detailed justification of the choice of criteria and a comparison to other approaches). While the term “check” might imply binary yes/no or present/absent comparators, I postulate that their operation and results are as differentiated and complex as the information-processing capacity of the respective organism allows. In many cases this consists of a continuous or graded appraisal on a scalar criterion and/or a multidimensional evaluation.[6] As will be argued in greater detail in the remainder of the chapter, it is suggested that the type and intensity of emotion that is elicited by a particular event depends on the profile of results of the appraisal process based on these SECs.

The Nature of the Stimulus Evaluation Checks

The SECs postulated in the most recent version of the model are organized in terms of four *appraisal objectives*. These objectives concern the major *types or classes of information* with respect to an object or event that an organism requires in order to prepare an appropriate reaction:

1. How relevant is this event for me? Does it directly affect me or my social reference group? (*relevance*)

2. What are the implications or consequences of this event and how do these affect my well-being and my immediate or long-term goals? (*implications*)
3. How well can I cope with or adjust to these consequences? (*coping potential*)
4. What is the significance of this event with respect to my self-concept and to social norms and values? (*normative significance*)

The checks that are suggested to be responsible for the production of this information are described in detail hereafter, grouped by the appraisal objectives.[7] It is important to stress that the outcomes of all of the SECs described here are always subjective and depend exclusively on the appraising individual's perception of and inference about the characteristics of an event. While under normal circumstances and for "reality-testing" individuals this subjective perception will bear some resemblance to the objective event characteristics, the two can diverge rather drastically in some cases (see Perrez & Reicherts, 1995). Furthermore, individual differences (see van Reekum & Scherer, 1997), transitory motivational states or moods (Forgas, 1991), and cultural values, group pressures, and the like can strongly influence the evaluation process (see Mesquita, Frijda, & Scherer, 1997; Scherer, 1997a, 1997b; and Manstead & Fischer and Mesquita & Ellsworth, this volume).

Relevance Detection

Organisms need to constantly scan external and internal stimulus input to check whether the occurence of stimulus events (or the lack of expected ones) requires deployment of attention, further information processing, and possibly adaptive reaction or whether the status quo can be maintained and ongoing activity pursued. Most important, given the constant barrage of stimulation, the organism must decide which stimuli are sufficiently relevant to warrant more extensive processing.

Novelty check. At the most primitive level of sensory-motor processing, any sudden stimulus (characterized by abrupt onset and relatively high intensity) is likely to be registered as novel and deserving attention (producing an orientation response; see Siddle & Lipp, 1997). Beyond this lowest level of novelty detection (which I will call *suddenness* detection; see also Tulving & Kroll, 1995), the assessment of novelty may vary greatly for different species, different individuals, and different situations and may depend on motivational state, prior experience with a stimulus, or expectation. One of the most important mechanisms might be schema matching to determine the degree of *familiarity* with the object or event (see Tulving et al., 1996, for the distinction between suddenness and familiarity detection). On a still higher level of processing, the novelty evaluation is likely to be based on complex estimates (based on an observation of regularities in the world) of the probability and predictability of the occurrence of a stimulus.

Intrinsic pleasantness check. The evaluation of whether a stimulus event is likely to result in pleasure or pain (in the widest sense) is so basic to many emotional responses that affective feeling is often equated with the positive or negative reaction toward a stimulus. In contrast, in the model advocated here, the intrinsic pleasantness evaluation is considered as separate from, and prior to, a positive goal/need con-duciveness check. In particular, it is suggested that the pleasantness or unpleasantness detected by the intrinsic pleasantness check is a feature of the stimulus and even though the preference may have been acquired, it is independent of the momentary state of the organism. In contrast, the positive evaluation of stimuli that help to reach goals or satisfy needs depends on the relationship between the significance of the stimulus and the organism's motivational state. In fact, intrinsic pleasantness is orthogonal to goal conduciveness, since something intrinsically pleasant (like chocolate cake) can be highly obstructive (if one is forced to eat three large pieces while trying to lose weight; see table 3 in Scherer, 1988a). The intrinsic pleasantness check determines the fundamental reaction or response of the organism: liking or pleasurable feelings, generally encouraging approach, versus dislike or aversion, leading to withdrawal or avoidance (in the most primitive form, a defense response; Vila & Fernandez, 1989).

Goal relevance check. This check establishes the relevance, pertinence, or importance of a stimulus or situation for the momentary hierarchy of goals/needs. A stimulus is relevant for an individual if it results in outcomes that affect major goals/ needs. Relevance is likely to vary continuously from low to high, depending on the number of goals/needs affected and/or their relative status in the hierarchy. For example, an event is much more relevant if it threatens one's livelihood or even one's survival than if it just endangers one's need to listen to a piece of music.[8]

Implication Assessment

This is a central appraisal objective since it determines to what extent a stimulus or situation is appraised as furthering or endangering an organism's survival and adaptation to a given environment, as well as satisfying its needs and attaining its goals. While the phenomena of motivation and goal-directed behavior are central to behavioral science, the present state of the field makes it rather difficult to specify the motivational constructs underlying appraisal more concretely. Even the terminology in this area is rather confusing: there is no consensus on the differential usage of terms such as *drives, needs, instincts, motives, goals, concerns*, and so on. In what follows the term *goal /need* will be used as shorthand for all of these motivational constructs.[9]

Causal attribution check. The organism will first attempt to attribute the causes of the event, in particular, to discern the agent that was responsible for its occurrence. In the case that the agent is an animate being, inference will also be made with respect to the motive or intention involved. The attribution processes used to gather this information can be quite complex (see Weiner, 1985a, for some of the factors involved). Obviously, the evaluation of the further evolution of the situation, in particular the probability of the outcomes and one's ability to deal with these, will greatly depend on the result of the attribution of agency and intention. For example, a student who received a failing grade in a final examination will appraise the situation very differently depending on whether he attributes the grade to a transcription error or to the professor's intent to sanction his lack of sufficient attention to the course work.

Outcome probability check. The central tenet of appraisal theory is that it is not the event itself but the perceived outcomes for the individual that determine the ensuing emotion (see also Lazarus and Roseman & Smith in this volume). In consequence, the individual needs to assess the likelihood or certainty with which certain consequences are to be expected. This is of particular importance in the case of *signal events* (e.g., a verbal threat) where both the probability of the signaled event occurring and its consequences are in doubt. But even in those cases in which an event has already happened, the actual consequences for the individual in terms of the likely outcomes need to be determined via probability estimates. For example, if a student fails an exam, some of the potential outcomes, for example, the reaction of the parents, can only be assessed in a probabilistic fashion.

Discrepancy from expectation check. The situation created by the event can be consistent or discrepant with the individual's expectation concerning that point in time or position in the action sequence leading up to a goal. For example, there would be some discrepancy in expectation if the father of the failed student gave him a present after learning of the exam results. The degree of discrepancy or consistency can be determined by the number of features or elements that fit the original expectation.[10]

Goal/need conduciveness check. Most important, the organism needs to check the conduciveness of a stimulus event to help attain one or several of the current goals/ needs. The consequences of acts or events can constitute the attainment of goals/ needs, or progress toward such attainment, or facilitation of further goal-directed action (see Oatley & Duncan, 1994). The more directly outcomes of events facilitate or help goal attainment, and the closer they propel the organism toward reaching a goal, the higher the conduciveness of an event. In other cases, results of events can be obstructive for goal attainment, by putting goal or need satisfaction out of reach, delaying its attainment, or requiring additional effort (see Srull & Wyer, 1986). This is the classic case of "frustration," the blocking of a goal-directed behavior sequence. Again, the obstruction can be more or less pronounced, depending on how severely goal attainment is hampered. In our failed examination example, the degree of obstructiveness is likely to depend on the role of the respective grade for completing the program and the difficulty of repeating the exam. Conduciveness is orthogonal to expectation—we can encounter highly conducive events that are very discrepant from pessimistic expectations or very obstructive events that are consistent with our worst premonitions (see table 4 in Scherer, 1988a).

Urgency check. Adaptive action in response to an event is particularly urgent when high-priority goals/needs are endangered and the organism has to resort to fight or flight and/or when it is likely that delaying a response will make matters worse. Urgency is also evaluated on a continuous scale: the more important the goals/needs and the greater the time pressure, the more urgent immediate action becomes. While any event evaluated as requiring an urgent response must be considered relevant, the reverse is not true. Urgency depends not only on the significance of an event for an organism's goal /need but also on temporal contingencies.

Coping Potential Determination

Successful coping with a stimulus event implies that the individual's concern with the eliciting event disappears and that the synchronized subsystems can be decoupled. It does not, however, necessarily require that the organism is able to reach its original goals. Coping can also consist of happily resigning oneself to a situation beyond one's control. For example, our failed student might decide that one can do very well in life without a university degree and turn to the stock exchange instead. The coping potential check determines which types of responses to an event are available to the organism and which consequences will affect the organism under each option. The net result of the evaluation on this check is the estimated degree of coping potential for the most promising response option available to the individual in this situation.

Control check. One important dimension is to what extent an event or its outcomes can be influenced or controlled by natural agents (i.e., people or animals). For example, while the behavior of a friend or the direction of an automobile are generally controllable up to a degree, the weather or the incidence of a terminal illness are usually not. In our example, the student would attribute little control if grades were routinely determined by a lottery. It is important to distinguish control from predictability as discussed earlier. Thus one may be able to predict the course of a hurricane with some degree of precision without being able to influence it. However, this does not work in reverse: generally control, in particular as far as the offset of a stimulus is concerned, implies predictability (see Mineka & Henderson, 1985, pp. 508–509, for a detailed discussion of this point).

Power check. If control is possible, coping potential depends on the power of the organism to exert control or to recruit others to help. With the help of the power check the organism evaluates the resources at its disposal to change contingencies and outcomes according to its interests. The sources of power can be manifold—physical strength, money, knowledge, or social attractiveness, among others (see French & Raven, 1959). In the case of the failed exam, the student might believe that having an uncle who presides over the university's endowment fund may bestow some relationship power that will help to get the grade changed. In the case of an obstructive event brought about by a conspecific aggressor or a predator, the comparison between the organism's estimate of its own power and the agent's perceived power is likely to decide between anger or fear and thus between fight or flight.[11]

The independence of the control and power checks needs to be strongly emphasized, since these two criteria are not always clearly distinguished in the literature, where "controllability" often seems to imply both aspects (see discussions in Garber & Seligman, 1980; Miller, 1981; Öhman, 1987). Control, in the sense that the term is used here, refers exclusively to the probability that an event can be prevented, or brought about, or have its consequences changed by a natural agent. Power, on the other hand, refers to the likelihood that the organism is actually able either by its own means or with the help of others to influence a potentially controllable event. A similar distinction has been suggested by Bandura (1977) in contrasting outcome expectation (contingency between response and outcome) and efficacy expectation (assumption that one's own response can produce the desired outcome).

Adjustment check. Organisms can adjust, adapt to, or live with the consequences of an event more or less well after all possible means of intervention have been used. It is particularly important to check how well one can adjust to the consequences of an event if the control and power checks yield the conclusion that it is not within one's power to change the outcomes. As mentioned already, the failed student might be able to live perfectly well with a terminal failing grade if he was convinced that his future should in any case be sought in the world of finance.

Normative Significance Evaluation

In socially living species it is not unimportant for the reaction of an organism to take into account how the majority of the other group members evaluate an action and to evaluate the significance of an emotion-producing event for its self-concept and self-esteem. Obviously, this appraisal objective is—by definition—only relevant in socially organized species capable of a self-concept and of a mental representation of sociocultural norms and values. This appraisal objective is served by two checks.

Internal standards check. This check evaluates the extent to which an action falls short of or exceeds internal standards such as one's personal self-ideal (desirable attributes) or internalized moral code (obligatory conduct). These can often be at variance with cultural or group norms, particularly in the case of conflicting role demands or incompatibility between the norms or demands of several reference groups or persons. For example, the failed student will react with a different emotion if he has an ideal self-concept of being a brilliant scholar in the respective field and having written a very creative essay for the exam in question, as compared to a self-ideal of financial wizard.

External standards check. Social organization in groups implies shared values and rules (norms) concerning status hierarchies, prerogatives, desirable outcomes, and acceptable and unacceptable behaviors. The existence and reinforcement of such norms depends on appropriate emotional reactions of group members to behavior-violating norms as well as to conforming behavior. The most severe sanction a group can use against a norm violator, short of actual aggression, is the display of emotional aversion and his relegation to the status of an outsider or a reject, thus depriving the individual of the positive emotional atmosphere of group contact. Therefore, in many cases, evaluating the significance of a particular action in terms of its social consequences is a necessary step before finalizing the result of the evaluation process and deciding on appropriate behavioral responses. This check evaluates to what extent an action is compatible with the perceived norms or demands of a salient reference group in terms of both desirable and obligatory conduct. Our student would appraise his failure quite differently on this check, depending on whether he applies the intellectual expectations and study performance standards of a group of junior scientists or that of the university football team. As in the case of the preceding SECs, the result for these checks is a value on a continuous dimension, indicating the degree of consistency of the action that is evaluated with external or internal standards.[12]

The Process of Appraisal

This enumeration of the SECs might give the impression that the significance of an object or event is just "checked" once. However, appraisal is not a one-shot affair. Very early, Lazarus (1966; Lazarus, Averill, & Opton, 1970) pointed out that appraisal is often followed by "reappraisal," serving to correct the evaluation results based on new information or more thorough processing. Organisms constantly scan their environment (and their internal state) to detect and reevaluate changes. Consequently, the current theory postulates that events or internal changes trigger cycles of appraisal running through the evaluation checks proposed here until the monitoring subsystem signals termination of or adjustment to the stimulation that originally elicited the appraisal episode.

This emphasis on the *process* of appraisal is supplemented by the claim that the SECs are processed in sequence, following a fixed order. While the SECs postulated in the sequential check theory of emotion differentiation correspond closely to what other appraisal theorists have postulated as appraisal criteria or dimensions (see also Scherer, 1988a, 1999b; Roseman & Smith and Schorr [a]; this volume), the sequence assumption is unique to the present theory. The SECs just described are expected to occur in a theoretically postulated sequence consisting of four stages in the appraisal process that correspond to the appraisal objectives described: (1) detection of stimulus characteristics that require attention deployment and further information processing; (2) assessment of the significance of the event (in particular, its implications and consequences) for the organism's goals and needs; (3) determination of the available coping potential; and (4) evaluation of the normative significance (in terms of values, norms, or standards) of the event and its aftermath for the self and its social surround.

Figure 5.1 shows a first attempt to model the architecture of the appraisal system and the underlying process as postulated by the theory. What are the arguments that could justify a sequence assumption? In terms of systems economy it seems useful to engage in expensive information processing only upon detection of a stimulus that is considered relevant for the organism and consequently requires attention. In consequence, *relevance detection* is considered to be a first selective filter a stimulus or event needs to pass to merit further processing. It is assumed that objects or events that surpass a certain threshold on novelty *or* intrinsic pleasantness/unpleasantness *or* goal /need relevance will pass this filter. Attention will be focused on the event, and further processing will ensue. It can be expected that hard-wired novelty and valence detectors as well as more controlled memory-based processes are involved in relevance detection.

Extensive further processing and preparation of behavioral reactions are useful only if the event actually concerns a goal or need of major importance or when a salient discrepancy with an expected state is detected. This requires that the nature of the event and all of its consequences, that is, the *implications* for the organism, be assessed. These are rarely obvious and may require logical operations, inferences, prob-abiliy assessments, and extrapolations into the future. Unless there are established schemata for the type of event, it is likely that most of the central nervous system (CNS) processing modules (memory, motivational hierarchy, reasoning) are involved. The causes and implications of the event and its cause need to be established *before* the organism's *coping potential* can be conclusively determined since the latter is always evaluated with respect to a specific demand (see Lazarus's insistence, 1991a, on the *transaction* between individual and event). Again, unless schematic processing is possible, many of the CNS processing modules, including the self-concept, will be involved in coping potential determination. Once all this information is in, the overall *significance* of an event and its consequences for the self and its normative/moral status can be evaluated.

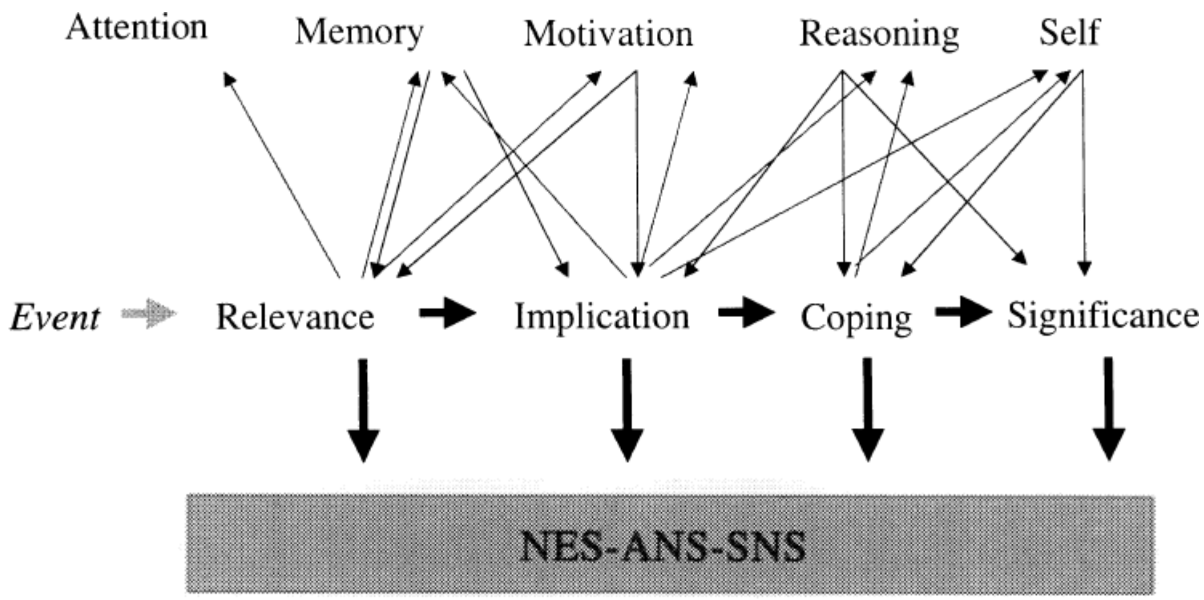

Figure 5.1. Afferent and efferent links of the elements in the appraisal process with associated cognitive structures and peripheral systems.

This thumbnail sketch of the appraisal process illustrates why I have originally postulated (Scherer, 1981b; 1984c) that the SECs are processed in a sequence, following the order described here. It seems defensible to argue that—for logical and economical reasons—the results of the earlier SECs need to be processed before later SECs can operate successfully, that is, yield a conclusive result.

This sequence assumption does not deny the existence of parallel processing. As shown in figure 5.2 (discussed in greater detail in the next section) all SECs are expected to be processed simultaneously, starting with relevance detection. However, the essential criterion for the sequence assumption is the point in time at which a particular check achieves *preliminary closure*, that is, yields a reasonably *definitive* result, one that warrants efferent commands to response modalities (the lozenges in figure 5.2; see detailed explanation hereafter). The sequence theory postulates that—for the reasons just outlined—on a macro level the result of a prior processing step (or check) must be in before the consecutive step (or check) can produce such a response-inducing result.

I have argued that the microgenetic unfolding of the emotion-antecedent appraisal processes parallels both phylogenetic and ontogenetic development in the differentiation of emotional states. Since the earlier SECs, particularly the novelty and the intrinsic pleasantness check, seem to be present in most animals as well as newborn humans, one could argue that these very low-level processing mechanisms take precedence as part of our hard-wired detection capacities and occur very rapidly after the occurence of a stimulus event. This should be particularly true for low-level novelty and intrinsic pleasantness detection. More complex evaluation mechanisms are successively developed at more advanced levels of phylogenetic and ontogenetic development, with natural selection operating in the direction of more sophisticated information-processing ability in phylogenesis and with maturation and learning increasing the individual's cognitive capacity in ontogenesis (see Scherer, 1984c, pp. 313–314; Scherer, Zentner, & Stern, submitted).

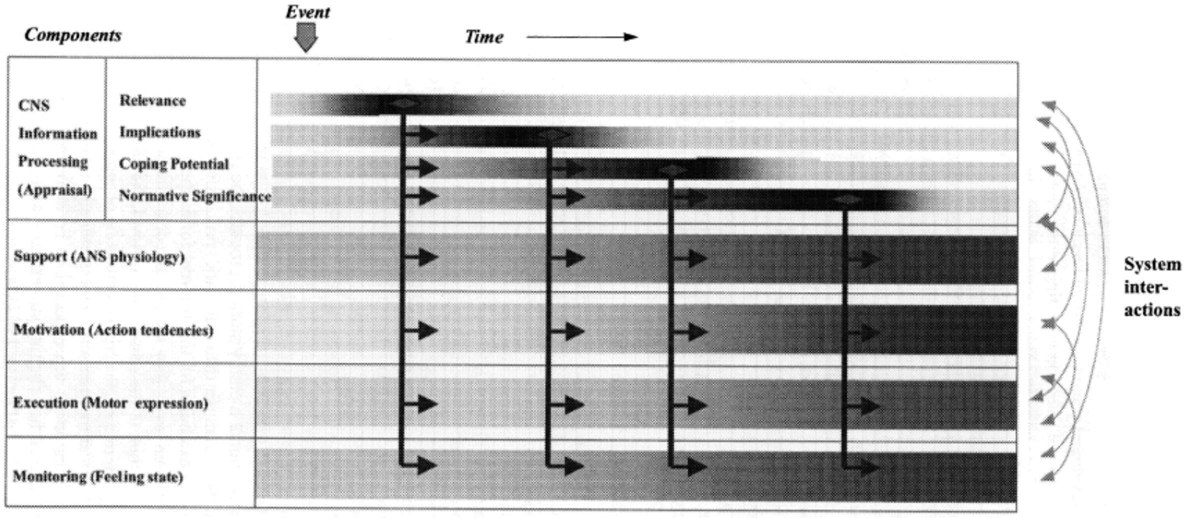

Figure 5.2. The component process model: Parallel processing with sequentially occurring efferent discharges following sufficiently conclusive results on evalution checks.

The assumption of sequential processing of the stimulus evaluation checks, and, in particular, the notion of a fixed order of the SECs, is often challenged. It is pointed out that the onset of an emotional reaction can occur extremely rapidly. However, the apparent speed of an emotional reaction to an event does not rule out a sequential model, given the rate with which these evaluations can occur (particularly when lower brain structures are involved). It is quite probable that massively distributed parallel processing, as emphasized by much of cognitive psychology and artificial intelligence (McClelland & Rumelhart, 1985), is at the root of the appraisal process. However, such parallel processes change rapidly over time, and it seems quite feasible to assume that the results of parallel processes for different evaluation criteria will be available at different times, given differential depth of processing (see the discussion of hierarchical processing in Minsky, 1985; Minsky & Papert, 1969).

Levels of Processing in Appraisal

The issue of the appraisal mechanisms and their role in emotion elicitation has been at the root of the controversy between Zajonc (1980, 1984a) and Lazarus (1984a, 1984b) on cognition-emotion interrelationships (see Schorr [a], this volume). In response to this controversy, Leventhal and Scherer (1987) pointed to the need to go beyond the semantic problem of defining emotion and cognition by specifying the details of emotion-antecedent information processing. They proposed a preliminary description of how the stimulus evaluation checks can occur on all three levels of the emotion-processing system postulated by Leventhal—the sensory-motor, the schematic, and the conceptual level (Leventhal, 1984; see also Buck, 1984b). On the lowest, the sensory-motor, level, checking occurs through largely innate feature detection and reflex systems that are specialized for the processing of specific stimulus patterns. On the schematic level, checking criteria are composed of schemata based on the learning history of the individual, which can be conceptualized as abstract representations of learned responses to specific stimulus patterns. On the conceptual level, finally, propositional memory storage provides the criteria for evaluation and conscious, and reflexive (rather than automatic) processing is used for checking.

Table 5.2 shows examples for the different forms the SECs can take depending on at which of the three levels they are processed. In line with the definition of the levels, it is assumed that, at the sensory-motor level, the checking mechanisms are mostly genetically determined, the criteria consisting of appropriate templates for pattern matching and similar mechanisms (correspondingly, sets of prewired autonomic-somatic response integrations are expected to be activated quasi-automatically). Prototypic unconditioned fear eliciting stimuli, such as are discussed in terms of “biological preparedness” (see Öhman, 1987), would be expected to be processed on this level. On the schematic level, the schemata forming the criteria for the SECs are expected to be largely based on social learning processes, and much of the processing at this level can be expected to occur in a fairly automatic fashion, outside of consciousness. It is likely that response integrations are stored along with the schema-eliciting criteria. On the conceptual level, the SECs are seen to be processed via highly cortical, propositional-symbolic mechanisms, requiring consciousness and involving cultural meaning systems. Here, responses are probably largely determined by volitional action. The multilevel, multicheck model proposed by Leventhal and Scherer (1987) is highly compatible with other multilevel processing theories pertinent to emotional processes that have been proposed in the literature (see van Reekum & Scherer, 1997, for a review). This model assumes that “higher” levels, allowing more sophisticated yet slower processing, are called into action only when the “lower” levels, relying on automatic and biologically prewired routines, cannot solve the problem. Furthermore, the different levels may continuously interact, producing top-down and bottom-up effects (see also Power & Dalgleish, 1997; van Reekum & Scherer, 1997).

Table 5.2. Levels of processing for stimulus evaluation checks

	Novelty	Pleasantness	Goal/need Conduciveness	Coping Potential	Norm/self Compatibility
1) Sensory-motor level	Sudden, intense stimulation	Innate preferences/ aversions	Basic needs	Available energy	(Empathic adaptation?)
2) Schematic level	Familiarity: schema matching	Learned preferences/ aversions	Acquired needs motives	Body schema	Self/social schemata
3) Conceptual level	Expectations: cause/effect, probability	Recalled, antici-pated, or derived positive-negative estimates	Conscious goals, plans	Problem-solving ability	Self ideal, moral evaluation

Source: Adapted from Leventhal and Scherer (1987, p. 17).

Recently, Smith and his collaborators have started to model these processes, drawing on concepts from cognitive psychology (Smith, Griner, Kirby, & Scott, 1966; see also Scherer, 1993a; Smith & Kirby, this volume). This attempt has motivated me to go beyond the suggestions in Leventhal and Scherer (1987) and try to construct a model that describes the process mechanisms that may underlie the sequential check theory in greater detail. Figure 5.3 shows the current state of that construction site. While these very preliminary suggestions will have to be taken with more than one grain of salt, they allow one to discern the direction that is envisaged for further development. Furthermore, this sketch of a process model suggests responses to questions such as: Are all subchecks carried out every time an emotion is generated? What is the interaction between the checks? What happens if "definitive " checking is too costly or too time-consuming?

I assume that the bulk of appraisal processing occurs in an information-processing system similar to the one described by Cowan (1988). The contents of the brief sensory storage (or sensory registers; see Karakas, 1997; Shiffrin & Atkinson, 1969) are processed, or "coded," by a range of procedures, from simple pattern matching to logical inference, based on schemata and representations that are activated in long-term memory. Consistent with the conceptualization by Leventhal and Scherer (1987) and the multiple path model suggested by LeDoux (1996), I assume that on a first pass, pertinent schemata are recruited in a largely automatic fashion to determine whether a satisfactory match (and, in consequence, a promising adaptational response) can be selected. In many cases, this is followed by controlled processing (regulated by a central executive processor) based on propositional content activated in long-term storage, giving rise to more elaborate evaluation and inference processes (see lower part of figure 5.3). I suggest that the results of both types of processing are stored in a set of *stimulation evaluation registers* (or *appraisal registers*), the contents of which are constantly updated. "Register" is used here in the classical sense of a rapidly accessible storage space for elementary information such as intermediate calculation results. It is suggested that there is one register for each of the checks (as shown by the enlarged representation of the registers in figure 5.3).

Given constant updating, each register will always hold a value that consists of the best available estimate of the respective aspect of the stimulus event. Thus, in parallel processing fashion, all checks will always be performed. This does not mean, however, that the content of the respective register will always have an impact on the adaptational response, that is, the type of emotion that will ensue (see hereafter).

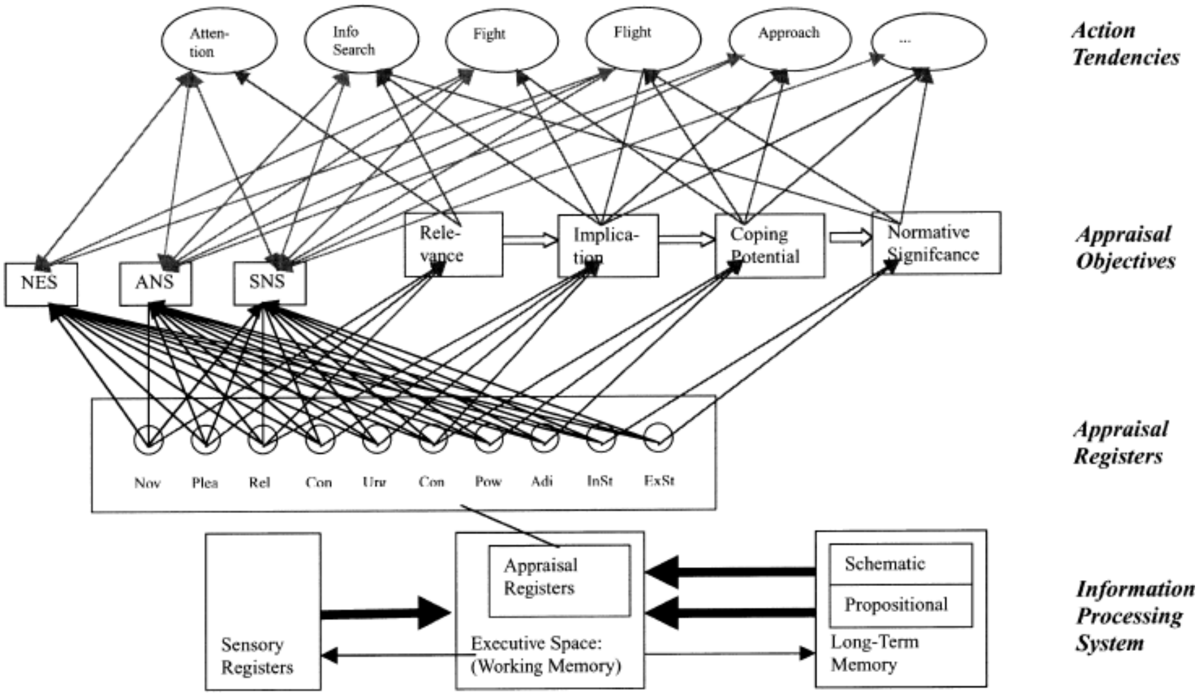

Figure 5.3. Sketch of the potential architecture of the appraisal process as part of a general information processing system, separately driving peripheral support systems and alternative action tendencies.

As suggested earlier, the SECs do the job of providing the four essential types of information required for action preparation: relevance, implication, coping, and normative significance. In consequence, the contents of the SEC registers need to be continuously combined or synthesized with respect to these classes of information.[13] In figure 5.3, this is demonstrated by the connections between the SEC registers and the boxes representing these information types. The specific integration or synthesis provided by the check registers for each of the information types will vary continuously as a function of information updating in the registers. It is expected that the different checks are integrated through weighting functions, giving them differential importance in the combination. These weighting functions may vary depending on the nature of the context (see Wehrle & Scherer, this volume).

A neural network architecture has been chosen for parts of figure 5.3 since it allows graphic representation of the connections and activation patterns in the model. The assumption is that the profile of synthesized information in the four major classes will activate potential response mechanisms (e.g., in the form of action tendencies; see Frijda & Zelenberg, this volume). A number of such action tendencies (as part of the executive subsystem) are shown in figure 5.3. As in a neural network model, one could assume that, depending on the profile of appraisal results, different action tendencies will be more or less strongly activated and will, in turn, activate the different parts of the support subsystem, the neuroendocrine system (NES), the autonomic nervous system (ANS), and the somatic nervous system (SNS). In contrast, the current version of the model, as shown in figure 5.2 and represented in figure 5.3 by the connections between check registers on the one hand and NES, ANS, and SNS on the other, assumes that there are direct connections of the SEC registers with these response modalities, independently of action tendencies.

However, as mentioned, it is assumed that such efferent effects will only occur if a minimal degree of closure (or definitiveness) of the evaluation of a specific check has been achieved (to avoid constant vascillation of the organism). The same is true for the level of synthesized information where, as shown in figure 5.1, it is expected that there will be sequential ordering of the moments of closure. One possibility of envisaging the mechanism for achieving closure that warrants efferent effects is to postulate markers of confidence or certainty for the register contents, combined with the temporal persistence of the same informational content in the register (which indicates that there has been no updating and that the information is stable). Thus, the contents of the registers will only have an influence on response patterning (and thus on the type of emotion) if the content is considered sufficiently reliable. It is likely that there is an interaction between appraised urgency and the confidence or reliability required for efferent effects—in cases where action is seen as urgently required, efferent effects may occur even at lower levels of appraisal reliability or certainty.

Given space limitation—as well as the preliminary state of the model in my mind—further details of the model remain to be specified elsewhere. Hereafter, the issue of the type of efferent effects one might expect on the basis of particular SEC results is explored.

The Componential Patterning Theory

In the component process model, emotion differentiation is predicted as the result of the net effect of all subsystem changes brought about by the outcome profile of the SEC sequence. On the basis of these assumptions a *componential patterning theory* is proposed that attempts to predict specific changes brought about by concrete patterns of SEC results.

The central assumption of the componential patterning theory is that the different organismic subsystems are highly interdependent and that changes in one subsystem will tend to elicit related changes in other subsystems. This process is not unidirectional. The postulates of general systems theory and neurophysiological evidence point to complex feedback and feedforward mechanisms between these subsystems. To take a simple example, an increase of CNS activation is likely to affect the SNS by increasing muscle tone, which, in turn, will feed back to the CNS and affect activation (Gellhorn, 1964). Yet in many cases the origin of a recursive chain of changing patterns is identifiable and can be traced to a specific subsystem. In the case of emotion, it is a reasonable assumption that the origin of a chain of changes is the information-processing subsystem and that the changes are produced by the evaluation of a specific event. Given the complexity of the interrelationships between the organismic subsystems, only the very first step of the sequence of reverberating changes will be discussed here—the effects of a specific evaluation check result on other subsystem states.

The basic tenet of the theory is that the result of each consecutive check will differentially and cumulatively affect the state of all other subsystems. This is shown in figure 5.2, which illustrates that—while there is continuous parallel processing in all components—there are moments in time (marked by the lozenges) when the evidence concerning a particular appraisal check is sufficiently strong to elicit efferent effects on the other components (the descending arrows). The darker coloration around the lozenges suggests more intense processing as preliminary closure on the information provided by the specific check is achieved. The suggestion is that these points of closure occur in a predetermined sequence, as argued earlier. The curved arrows at the right of the processing "box" indicate the pervasive interactions between the different components, including, most important, the possibility that the cognitive appraisal processes can be affected by changes in the other components.

The componential patterning model claims that (1) the outcome of each SEC changes the state of all other subsystems and (2) the changes produced by the result of a preceding SEC are modified by that of a consequent SEC. Let us take an example: the detection of a novel, unexpected stimulus by the novelty check will produce (1) an orientation response in the support system (e.g. heart rate decrease, skin conductance increase); (2) postural changes in the motivation (or action tendency) system (focusing the sensory reception areas toward the novel stimulus); (3) changes in goal priority assignment in the executive subsystem (attempting to deal with a potential emergency); and (4) alertness and attention changes in the monitor subsystem. When, milliseconds later, the next check, the intrinsic pleasantness check, reaches sufficient closure to determine that the novel stimulus is unpleasant, the efferent effects of this result will again affect the state of all other subsystems and thus modify the changes that have already been produced by the novelty check. For example, an unpleasant evaluation might produce the following changes: (1) a defense response in the support system (heart rate increase, etc.); (2) an avoidance tendency in the executive subsystem; (3) motor behavior to turn the body away from the unpleasant stimulation (thus reducing intake of stimulation in the action system; and (4) a negative subjective feeling in the monitor system. Similarly, all of the following checks will change the states of all other subsystems and will thus further modify the preceding changes.

It is important to note that each SEC result modifies the preceding changes. This has the effect that the patterning of the component states is specific to the unique evaluation "history" of the respective stimulus. For example, an unpleasant odor that was expected will yield a componential patterning that is different from an unexpected one since the changes produced by the preceding step are different. Thus in addition to specifying particular change patterns for the results of the SECs, the component patterning theory emphasizes the unique patterning of the emotion component states due to the particular sequence of modifications following the SEC results. In other words, the SECs and the effects of their results on the other subsystems are not independent of each other; rather, each preceding SEC result and the change produced by it "sets the scene" for the effects of the following SECs result. Looking at it in another way, specific "patterns" of component states (such as those that seem to characterize emotions like anger or fear) can only occur if there is a series of specific SEC results where each SEC adds a particular modification or "added value" in a complex sequential interaction (see Scherer, 1987a, for more detail).

It is beyond the scope of this chapter to provide a detailed discussion of the predictions for the effects of specific SEC results on the executive, support, and action systems in the component process model. These predictions follow directly from the functional approach adopted in the component process model, both in terms of the general functions of emotion as described at the beginning of the chapter and in terms of the specific functions of each SEC, as discussed. In particular, specific motivational and behavioral tendencies are expected to be activated in the executive subsystem in order to serve the specific requirements for the adaptive response demanded by a particular SEC result. For socially living species, adaptive responses are required not only in terms of the internal regulation of the organism and motor action for instrumental purposes (organismic functions) but also with respect to interaction and communication with conspecifics (social functions).

Table 5.3 (reproduced from Scherer, 1987a) presents the hypotheses concerning detailed response patterns for the CNS, the NES, the ANS, and the SNS. For the sake of brevity not all of the checks are listed, and in some cases only major combinations of check results are used. The origins of the predictions shown in table 5.3 are discussed in greater detail in Scherer (1987a). While these predictions will have to be constantly revised in line with research progress, the table is reproduced here in its original form for the sake of illustration. With my collaborators, I am currently engaged in adapting the predictions to the evolution of research evidence, both from our own and other laboratories. The first results of these efforts have been presented for facial expression (see Wehrle, Kaiser, Schmidt, & Scherer, 2000, Kaiser & Wehrle, this volume) and physiological and vocal predictions (Johnstone, van Reekum, & Scherer, this volume). However, it seems too early to take stock of these modifications in order to present a major update of table 5.3.

The general assumption is that as a threshold of closure or definitiveness of a decision for a criterion or check is reached (the lozenge symbols in figure 5.2), there will be efferent effects on the other components of emotion (the response modalities), according to the predictions described earlier. The cumulative effects of this process are expected to constitute emotion-specific response profiles, as described hereafter.

Predicting SEC Profiles for Modal Emotions

So far, in discussing the differentiation of emotional states, I have rarely used the standard emotion words that generally characterize discussions of affect and emotion—anger, fear, joy, sadness, disgust, and the like. Contrary to discrete emotion theories (Ekman, 1984, 1992a; Izard, 1977, 1993; Tomkins, 1984), the component process model does not share the assumption of a limited number of innate, hard-wired affect programs that mix or blend with each other in order to produce the enormous variety of different emotional states. Rather, the emotion process is considered as a continuously fluctuating pattern of change in several organismic subsystems, yielding an extraordinarily large number of different emotions.

However, there is no denying that there are some major patterns of adaptation in the life of animate organisms that reflect frequently recurring patterns of environmental evaluation results. All organisms, at all stages of ontogenetic development, encounter blocks to need satisfaction or goal achievement at least some of the time. Thus frustration in a very general sense is universal and ubiquitous. Equally universal are the two major reaction patterns—fight and flight. Consequently, it is not surprising that the emotional states that often elicit these behaviors, anger and fear, respectively, seem universal and are present in many species. In terms of the model proposed here, it is highly likely that, if one were to compile a frequency distribution of SEC patterns, some combinations of SEC results would be found to be very frequently encountered by many types of organisms, giving rise to specific, reccurring patterns of state changes. It has been suggested (Scherer, 1984a, 1994b) that the term *modal emotions* be used for the states resulting from these predominant SEC outcomes due to general conditions of life, constraints of social organization, and similarity of innate equipment. In most languages of the world, these modal emotions have been labeled with a short verbal expression, mostly a single word.

Table 5.3. Component patterning theory predictions for CNS, NES, ANS, and SNS changes following major SEC outcomes

NOVELTY CHECK		
	Novel:	Not novel:
GENERAL	orienting response, interruption of ongoing activity	No change
CNS	EEG alpha blocking, P300 component in evoked cortical potential	
NES	corticosteroid secretion	
ANS	brief inhibition of respiration followed by sudden inhalation, HR deceleration, vasomotor changes, skin conductance responses, pupillary dilatation	
SNS	local tonus changes	
FACE	AUs 1,2 (brows up), 5 (lids up), or 4, 7 (frown, scanning), 26 (jaw drop, open mouth), 38 (open nostrils); gaze directed	
VOICE	interruption of phonation, ingressive (fricative) sound with glottal stop (noise-like spectrum)	
BODY	interruption of ongoing instrumental action, raising head, straightening posture	
INTRINSIC PLEASANTNESS CHECK		
	Pleasant:	Unpleasant:
GENERAL	sensitization of sensorium	defense response, desensitization of sensorium
CNS	—	EEG alpha blocking
NES	—	corticosteroid secretion
ANS	inhalation, HR deceleration, increase in glandular secretions, particularly salivation, pupillary dilatation	HR acceleration, increase in SC level, decrease in salivation, pupillary constriction
SNS	local tonus changes	slight tonus increase
FACE	AUs 5 (lids up), 26 (jaw drop, open mouth), 38 (open nostrils) or 12 (lip corners pulled upwards), 25 (lips part); gaze directed	AUs 4 (brow lowering), 7 (lid tightening), eye closing (possibly intermittent), 9 (nose wrinkling), 10 (upper lip raising), 15 (lip corner depression), 17 (chin raise), 24 (lip press), 39 (nostril compression); or 16 (lower lip depressed), 19 (tongue thrust), 25 (lips part), 26 (jaw drop); gaze aversion
VOICE	faucal and pharyngeal expansion, relaxation of tract walls, vocal tract shortened due to AU 25 action (increase in low frequency energy, F1 falling, slightly broader F1 bandwidth, velopharyngeal nasality, resonances raised—"wide voice")	faucal and pharyngeal constriction, tensing of tract walls, vocal tract shortened due to AU 15 action (more high frequency energy, F1 rising, F2 and F3 falling, narrow F1 band-width, laryngopharyngeal nasality, resonances raised/"narrow voice")
BODY	centripetal hand and arm movements, expanding posture, approach locomotion	centrifugal hand and arm movements, hands covering orifices, shrinking posture, avoidance locomotion
GOAL/NEED CONDUCIVENESS CHECK		
	Relevant and consistent:	Relevant and discrepant:
GENERAL	trophotropic shift, rest and recovery	ergotropic shift, preparation for action
CNS		EEG alpha blocking
NES	EEG synchronization	corticosteroid and catecholamine, particularly adrenaline secretion
ANS	decrease in respiration rate, slight HR decrease, vasodilatation in sexual organs, increase in glandular secretion, bronchial constriction, increase in gastrointestinal motility, relaxation of sphincters	deeper and faster respiration, increase in HR and heart stroke volume, vasoconstriction in skin, gastrointestinal tract, and sexual organs, vasodilatation in heart and striated musculature, increase of glucose and free fatty acids in blood, decreased gastrointestinal motility, sphincter contraction, bronchial dilatation, contraction of m. arrectores pilorum, decrease of glandular secretion, increase in SC level, pupillary dilatation
SNS	decrease in general muscle tone	increased muscular tonus
FACE	relaxation of facial muscle tone	AUs 4 (brow lowerer, frown), 7 (lids tighten), 23 (lips tighten), 17 (chin raising); gaze directed
VOICE	overall relaxation of vocal apparatus (F0 at lower end of range, low-to-moderate amplitude, balanced resonance with slight decrease in high-frequency energy—"relaxed voice")	overall tensing of vocal apparatus (F0 and amplitude increase, jitter and shimmer, increase in high frequency energy, narrow F1 bandwidth, pronounced formant frequency differences—"tense voice")

BODY	comfort and rest positions	strong tonus, task-dependent instrumental actions
IF CONDUCIVE TO GOAL	trophotropic shift plus elements from pleasantness response	ergotropic shift with elements of pleasantness response
IF OBSTRUCTIVE TO GOAL	trophotropic shift plus elements from unpleasantness response	ergotropic shift with elements of unpleasantness response
COPING POTENTIAL CHECK		
	No control:	
GENERAL	Trophotropic dominance (possibility of ergotropic rebound?)	
CNS	Cortical alpha rhythm	
NES	high level corticosteroid release, decrease in catecholamine, particularly noradrenaline secretion	
ANS	decrease in respiration rate and depth, HR decrease, increase in glandular secretion, particularly tear glands, bronchial constriction, increase in gastro-intestinal motility, relaxation of sphincters	
SNS	hypotonus of the musculature	
FACE:	hypotonus of facial musculature, AUs 15 (lip corner depression), 25 (lips parting), 26 (jaw dropping), 41 (lids drooping), 43 (eyes closed); if tears 1 (inner brow raise), and 4 (brow lowered) gaze aversion	
VOICE	hypotonus of vocal apparatus (low F0 and restricted F0 range, low amplitude, weak pulses, very low high-frequency energy, spectral noise, format frequencies tending toward neutral setting, broad F1 bandwidth — "lax voice"	
BODY	few and slowed movements, slumped posture	
	Control and high power:	Control and low power:
GENERAL	Shift toward ergotropic-trophotropic balance	extreme ergotropic dominance
CNS	EEG alpha blocking	EEG alpha blocking
NES	Increase of noradrenaline component in catecholamine secretion, decrease in corticosteroid secretion	strong adrenaline preponderance in catecholamine secretion, strong corticosteroid secretion
ANS	increase in depth of respiration, slight heart rate decrease, increase in systolic and diastolic blood pressure, changes in regional blood flow, increased flow to head, chest, and hands (reddening, increased skin temperature in upper torso), increase in free fatty acids in blood (thermogenesis), some gastric motility, pupillary constriction	faster and more irregular respiration, strong increase in HR and heart stroke volume, increase in systolic and decrease in diastolic blood pressure, increase in pulse volume amplitude, vasoconstriction in skin (pallor, decreased skin temperature), gastrointestinal tract, and sexual organs, increase in blood flow to striated musculature, increase of glucose and free fatty acids in blood, decreased gastrointestinal motility, sphincter contraction, tracheobronchial relaxation, contraction of m. arrectores pilorum, decrease of glandular secretion, secretion of sweat (increase in SC level), pupillary dilatation
SNS	balanced tone, tension increase in head and neck	hypertonus, particularly in locomotor areas, trembling
FACE	AUs 4, 5 (eyebrows contracted, eyes widened), or 7 (lids tightened, eyes narrowed), 23, 25 (lips tight and parted, bared teeth), or 23, 24 (lips tight, pressed together), 38 (nostril dilation); stare	AUs 1,2,5 (brow and lid raising), 26 (jaw drop), 20 and platysma (mouth stretch and corner retraction), 38 (nostril dilation); switching between gaze direction and aversion
VOICE	Chest register phonation (low F0, high amplitude, strong energy in entire frequency range—"full voice")	head register phonation (raised F0, widely spaced harmonics with relatively low energy—"thin voice")
BODY	agonistic hand/arm movements, erect posture, body lean forward, approach locomotion	protective hand/arm movements, fast locomotion or freezing
NORM/SELF COMPATIBILITY CHECK		
	External/internal standards surpassed:	External/internal standards violated:
	Ergotropic shift plus elements of pleasantness and high power response	Ergotropic shift plus elements of unpleasantness and low power response (peripheral blood flow to face, blushing; body movements: active avoidance of communicative contact)

AU: action unit; HR: heart rate; SC: skin conductance

Since the component process model assumes that emotion differentiation is produced by the results of the SEC sequence, it follows that the latter should also be the key to predicting modal emotions. Table 5.4 shows the predicted profiles of the antecedent SEC result patterns expected to produce such modal emotions. In the first version of this table (Scherer, 1981b) the predictions made were based on a theoretical analysis. Since then they have been continuously refined on the basis of our own research and relevant work in other laboratories that appeared in recent years (see Scherer, 1999a). Table 5.4 presents the most recent version of these predictions.

For each of the SECs, a graded scale of typical result alternatives is assumed and both the polarity and the grading are used for the predictions. The term “open“ indicates that many different results of a particular check are compatible with the occurence of the respective modal emotion or that the check may be irrelevant for that emotion. The fairly high frequency of “open” entries can be interpreted as the basis for the emotion variants within a modal emotion family.

Similarly, emotions that are closely related with respect to the basic structure of the antecedent situation may be quite different qualitatively because of grading (or intensity) differences in the SEC results. This is true, for example, for the distinction between hot and cold anger or between worry and fear. The failure to distinguish between such related states may be the reason for the difficulty of replicating results in research on emotional responses since different investigators may have used similar-sounding labels for qualitatively rather different emotional states (see Banse & Scherer, 1996). By combining the predictions in tables 5.3 and 5.4, specific response profiles for vocal and facial expression, as well as ANS response patterns, have been predicted for these modal emotions (see Scherer, 1986a, 1987a, 1992c).

Empirical Evidence for the Sequential Check Theory of Emotion Differentiation

In this section, the empirical evidence that has been produced for the model presented here will be reviewed.

Similarity Analysis of Emotion Labels

Using multidimensional scaling (MDS) and cluster analysis of similarity judgments between 235 German emotion adjectives, we demonstrated that major appraisal criteria are reflected in the hierarchical organization of the semantic space formed by these emotion terms (Scherer, 1984a). Further analyses suggested a tetraeder model of emotional meaning that can be linked to the operation of central SECs (Gehm & Scherer, 1988b).

Table 5.4. Predicted appraisal patterns for some major modal emotions

	ENJ/HAP	ELA/JOY	DISP/DISG	CON/SCO	SAD/DEJ	DESPAIR	ANX/WOR
CRITERION							
Relevance							
Novelty							
Suddenness	low	high/med	open	open	low	high	low
Familiarity	open	open	low	open	low	very low	open
Predictability	medium	low	low	open	open	low	open
Intrinsic pleasantness	high	open	very low	open	open	open	open
Goal/need relevance	medium	high	low	low	high	high	medium
Implication							
Cause: agent	open	open	open	other	open	oth/nat	oth/nat
Cause: motive	intent	cha/int	open	intent	cha/neg	cha/neg	open
Outcome probability	very high	very high	very high	high	very high	very high	medium
Discrepancy from expectation	consonant	open	open	open	open	dissonant	open
Conduciveness	high	very high	open	open	obstruct	obstruct	obstruct
Urgency	very low	low	medium	low	low	high	medium
Coping potential							
Control	open	open	open	high	very low	very low	open
Power	open	open	open	low	very low	very low	low
Adjustment	high	medium	open	high	medium	very low	medium
Normative significance							
Internal standards compatibility	open	open	open	very low	open	open	open
External standards compatibility	open	open	open	very low	open	open	open

	FEAR	IRR/COA	RAG/HOA	BOR/IND	SHAME	GUILT	PRIDE
CRITERION							
Relevance							
Novelty							
Suddenness	high	low	high	very low	low	open	open
Familiarity	low	open	low	high	open	open	open
Predictability	low	medium	low	very high	open	open	open
Intrinsic pleasantness	low	open	open	open	open	open	open
Goal/need relevance	high	medium	high	low	high	high	high
Implications							
Cause: Agent	oth/nat	open	other	open	self	self	self
Cause: Motive	open	int/neg	intent	open	int/neg	intent	intent
Outcome probability	high	very high	very high	very high	very high	very high	very high
Discrepancy from expectation	dissonant	open	dissonant	consonant	open	open	open
Conduciveness	obstruct	obstruct	obstruct	open	open	high	high
Urgency	very high	medium	high	low	high	medium	low
Coping potential							
Control	open	high	high	medium	open	open	open
Power	very low	medium	high	medium	open	open	open
Adjustment	low	high	high	high	medium	medium	high
Normative significance							
Internal standards	open	open	open	open	very low	very low	very high
External standards	open	low	low	open	open	very low	high

ENJ/HAP: enjoyment/happiness; ELA/JOY: elation/joy; DISP/DISG: displeasure/disgust; CON/SCO: contempt/scorn; SAD/DEJ: sadness/dejection; ANX/WORR: anxiety/worry; IRR/ COA: irritation/cold anger; RAG/HOA: rage/hot anger; BOR/IND: boredom /indifference; med: medium; oth: other; nat: natural; int: intentional: cha: chance; neg: negligence.

Appraisal Ratings of Recalled Emotion Experiences

Based on the results of earlier crosscultural studies of emotional experience (Scherer, Wallbott, & Summerfield, 1986), we used verbal report of recalled emotional experience in a study with approximately 3000 respondents in 37 countries all over the world. Respondents were asked to describe recent situations that had provoked fear, anger, sadness, joy, disgust, shame, and guilt, respectively. In addition to reporting the duration and intensity of the subjective feeling and the nature of the verbal, nonverbal, and physiological responses, they were asked a number of questions on how they had appraised the event (the questions having been formulated in accordance with the SEC model). A first, nonmetrical analysis of a partial data set from this study showed that it was possible to predict the type of emotion on the basis of the appraisal questions (Gehm & Scherer, 1988b). More recently, the complete data set was analyzed. Table 5.5 (adapted from Scherer, 1997a) contrasts the predictions with the results (expressed in standard score profiles; indicating the direction and the importance of each check for a particular emotion). While it is apparent that this method is much too imprecise to allow a detailed test of the predictions made in table 5.4, the results generally supported the predictions (although some of the SECs had less of an impact than had been theoretically predicted, especially in the case of fear). Since, given the scope of the study, the questions had to be extremely brief and simple, it is reasonable to assume that more comprehensive questions, possibly using a direct interviewing procedure, may provide a satisfactory empirical test of the predictions.[14]

Table 5.5. Predictions and Z-score profiles for emotion-specific appraisals from a study of 37 countries

	Joy	Fear	Anger	Sadness	Disgust	Shame	Guilt
Expectedness	open 0.64 e	low −0.12 bc	open −0.21 a	open 0.03 d	open −0.19 a	open −0.13 ab	open −0.01 cd
Unpleasantness	low −2.00 a	high 0.34 cd	open 0.41 e	open 0.36 d	very high 0.40 e	open 0.24 b	open 0.25 b
Goal hindrance	very low −1.18 a	high 0.12 b	high 0.37 c	high 0.30 c	open 0.13 b	open 0.13 b	low 0.14 b
External causation	open −0.13 c	external 0.23 e	external 0.12 d	open 0.47 f	external 0.27 e	internal −0.41 b	internal −0.55 a
Coping potential	medium 0.44 c	very low −0.29 a	high 0.07 b	low −0.35 a	open 0.02 b	open 0.02 b	open 0.08 b
Unfairness	np −0.72 a	np 0.00 c	np 0.58 f	np 0.11 d	np 0.27 e	np −0.13 b	np −0.11 b
Immorality	open −0.63 a	open −0.05 c	high 0.29 e	open −0.12 b	open 0.29 e	open 0.05 d	very high 0.16 d
Self-consistency	open 1.18 d	open −0.06 c	low −0.09 c	open −0.17 b	open −0.02 c	very low −0.41 a	very low −0.43 a

Emotions that differ with respect to the trailing letters are significantly different from each other in a *posthoc* comparison. It should be noted that the predictions reported here are based on an earlier version of the theory and are not directly comparable to those in Table 5.4 . Np = not predicted.

Source: Reproduced and modified from Scherer (1997a, pp. 138–139).

In general, then, studies in which respondents have been asked to recall emotional incidents and to report on their inferred appraisal processes have provided some support for the theory presented here (as has been the case for other appraisal theories; see Scherer, 1999a; Roseman & Smith, this volume

). What seems important for the future is to systematically compare the sequential check theory with other approaches. Such comparative testing has been relatively rare (but see Mauro, Sato, & Tucker, 1992; Roseman, Spindel, & Jose, 1990; Scherer, 1999b).

In other studies, a computerized expert system has been used to predict the emotions people have felt on the basis of verbally reported appraisal profiles (Scherer, 1993b). The task consists in remembering a strong emotion one has experienced recently and answering a series of questions (based on the SECs described earlier), posed by the computer, concerning the evaluation of the situation that has elicited the respective emotion. Once all questions are answered, the computer suggests a label to describe the emotion one has experienced (based on the predictions shown in table 5.4) and the respondent can indicate whether he or she considers this to be a correct diagnosis or not. The data in the original study (Scherer, 1993b) yielded a correct diagnosis in 77.9% of all cases. However, this impressive mean accuracy hides a rather poor performance for anxiety and fear. More than 500 participants in several countries have been studied with the expert system since. The analyses (using the GATE system; see Wehrle & Scherer, this volume) confirm the ability of the system (based on the theory outlined here) to predict emotion labels for actually experienced situations with better than chance accuracy (although the accuracy percentages vary from study to study and fear remains a problem). Edwards (1998) has confirmed and refined these results. In addition, he demonstrated that intensity can be predicted on the basis of the SECs but that the relative amount of the contribution to the variance in intensity made by different SECs varies over emotions.

ANS, Vocal, and Facial Response Predictions for Modal Emotions

Scherer (1986a) produced a number of concrete predictions for the acoustic characteristics to be expected as vocal manifestations for 14 modal emotions. In a large study using portrayals by professional theatre actors, Banse & Scherer (1996) tested these predictions. While the results showed that a number of predictions might have to be modified, a rather large percentage of the hypotheses was confirmed, thus supporting the general approach and many of the specific predictions. Currently, our research group is attempting to replicate these results in a study with induced rather than portrayed emotional states (Johnstone, 1996; see also Johnstone, van Reekum, & Scherer, this volume).

As for facial patterning, Scherer (1987a, 1992c) has predicted patterns of facial expressions (described objectively in the form of action unit configurations using the Facial Action Coding System; Ekman & Friesen, 1978). Wehrle et al. (2000) have used facial synthesis to test these predictions. The synthesized facial expressions that were based on the theoretical predictions of the sequential check model were recognized with much better than chance accuracy. In addition, the pattern of errors were similar to synthetic expressions modeled after photographs of well-recognized posed expressions. These results strongly encourage further research in this direction.

The predictions on facial patterning have also been tested with natural facial expressions filmed in the context of experimental computer games that allow manipulation of the appraisal processes (Wehrle et al., 2000; see also Kaiser & Wehrle, this volume). Again, the data generally support the predictions on facial patterning as based on the sequential check theory of appraisal outlined earlier (Schmidt, 1998). However, the empirical data also indicate many points where the predictions will have to be refined.

With respect to the predictions of physiological patterning in the componential process model, our research group has studied peripheral physiological responses to manipulated appraisal responses (in a computer game) for the dimensions of intrinsic pleasantness, goal/need conduciveness, and coping potential. While there are no significant effects for the intrinsic pleasantness manipulation, some of the results for the goal/need conduciveness and coping potential check manipulation are consistent with the predictions of the componential patterning model (Banse, Etter, van Reekum, & Scherer, 1996; van Reekum, Banse, Johnstone, Scherer, et al., submitted; Johnstone, van Reekum, & Scherer, 1999; van Reekum, Johnstone, & Scherer, 1997).

Empirical Investigation of the Sequence Hypothesis

Given (1) the difficulty of producing as well as assessing different SEC results and (2) the speed of change in the subsystems concerned, empirical testing of the sequence hypothesis requires a methodology that remains to be developed. While waiting for the development of such methods (see Scherer, 1993a), I used a research paradigm developed in cognitive psychology—the accuracy-speed tradeoff paradigm—to study the sequence issue (Scherer, 1999b). In this study, participants received information about an emotion-eliciting situation in the form of information elements structured according to the SECs. Their task was to recognize the emotion described by optimizing accuracy, decision time, and amount of information requested. The hypothesis that the emotions would be judged faster and more accurately when the information was provided in the theoretically predicted rather than in a random sequence was confirmed.

Outlook

The qualification of the model presented here as a “process model” applies in more than one sense—as a theoretical edifice it has been under construction for a good number of years and is likely to remain that way. This chapter provides an overview of the state of the model at this point in time. Since many of the changes introduced in this chapter are recent, many issues remain to be resolved. However, quite independently of the intricacies of terminology and conceptualizations, which makes theory building such an arduous labor, it has been encouraging to see that appraisal theory is a powerful generator for empirical research. Even more important, experience has shown that the data thus generated exert a strong pressure toward the constant modification of the theory. I submit that this constant interchange between data and theory is a good sign for the health of our field and justifies hope that we will soon better understand the elicitation and differentiation of emotion.